

An Empirical Analysis of Hallucination Behavior in LLMs in the Medical Domain

Aidan Perez

MS in AI, Boston University
aperez00@bu.edu

Kaveri Goel

MS in AI, Boston University
kvrgl@bu.edu

Dang Dinh

MS in DS, Boston University
dddinh@bu.edu

Ying Huang

MS in CS, Boston University
yinghy@bu.edu

Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in generating fluent, contextually relevant text. However, they frequently produce hallucinations — outputs that appear plausible but are factually incorrect, unsupported, or fabricated. This problem is especially critical in high-stakes domains such as medicine, where incorrect information can have serious real-world consequences. We present a controlled empirical study of hallucination behavior across three model families (Llama-3-8B, BioMistral-7B, and BioGPT) on two medical QA benchmarks (MedQA and PubMedQA), evaluating four intervention conditions: zero-shot prompting, chain-of-thought (CoT) prompting, retrieval-augmented generation (RAG), and supervised fine-tuning (SFT). Our results show that larger instruction-tuned models benefit most from RAG, CoT prompting consistently degrades performance relative to zero-shot due to output noise and parse failures, and fine-tuning improves BioGPT accuracy by 6.66 percentage points while eliminating parse failures entirely. Combining fine-tuning with RAG, however, leads to catastrophic output degradation, suggesting that task-specific fine-tuning and retrieval augmentation are incompatible in their current forms for this model.

1 Introduction

Large Language Models (LLMs) have revolutionized natural language processing by demonstrating unprecedented capabilities in generating coherent, contextually relevant text. However, a significant challenge is their propensity to produce hallucinations — outputs that may be fluent and plausible but are factually incorrect, unsupported by evidence, or entirely fabricated. This issue is particularly concerning in medicine, where incorrect information can lead to serious consequences for patient care and safety.

Despite the growing body of research on hallucination in general NLP contexts, relatively little empirical work systematically characterizes hallucination behavior in medical LLMs, or directly compares the effectiveness of inference-time versus training-time mitigation strategies. In this paper, we address this gap by evaluating multiple LLMs on standardized medical QA benchmarks under four experimental conditions: zero-shot prompting, chain-of-thought prompting, retrieval-augmented generation, and supervised fine-tuning. By systematically comparing these interventions, we aim to answer not only how often LLMs hallucinate on medical questions, but which type of intervention works best — and whether combining them yields further gains.

2 Related Work

2.1 Hallucination in Large Language Models

Hallucination is a widely recognized problem in LLMs, where a model generates fluent and plausible text that is factually incorrect, unsupported by the input, or inconsistent with external knowledge. Prior work has categorized hallucinations into intrinsic hallucinations, which contradict the source context, and extrinsic hallucinations, which introduce information not supported by the provided evidence (Ji et al., 2023). Hallucination has also been studied in abstractive summarization, where models often generate coherent but unfaithful content (Maynez et al., 2020). These findings are relevant to medical LLMs because medical responses require both factual accuracy and faithfulness to clinical context.

2.2 Medical Question Answering and Clinical Reasoning

Recent work has shown that LLMs can achieve strong performance on medical QA benchmarks such as MedQA, MedMCQA, and PubMedQA,

but remain vulnerable to confident errors on complex clinical reasoning tasks (Liévin et al., 2024). Medical benchmarks provide a useful setting for studying hallucination because they offer verifiable ground-truth answers. Prior studies have also developed medical hallucination benchmarks, such as Med-HALT, which evaluates reasoning-based and memory-based hallucination in medical LLMs (Pal et al., 2023).

2.3 Chain-of-Thought Prompting

Chain-of-thought (CoT) prompting encourages a model to generate intermediate reasoning steps before producing a final answer (Wei et al., 2022). While CoT has been shown to improve reasoning on complex tasks, it can also produce longer generations that introduce additional opportunities for error. In medical QA, a model may generate a fluent but incorrect reasoning chain before giving its final answer, and CoT outputs may be harder to parse reliably (Wang et al., 2023).

2.4 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) combines a generative model with an external retrieval component, allowing the model to condition its output on retrieved documents rather than relying only on parametric knowledge (Lewis et al., 2020). In the medical domain, retrieval can help ground model outputs in biomedical abstracts or clinical documents, potentially reducing extrinsic hallucination. However, RAG does not fully eliminate hallucination, as a model may still misinterpret retrieved evidence or fail to align its answer with the retrieved context.

2.5 Fine-Tuning for Medical NLP

Supervised fine-tuning (SFT) on domain-specific data has been shown to improve model performance on medical tasks (Singhal et al., 2023). Domain-specific pretraining, as in BioGPT (Luo et al., 2022) and BioMistral (Labrak et al., 2024), has also been proposed as a way to reduce hallucination by internalizing medical knowledge during training. However, the relative effectiveness of fine-tuning versus inference-time interventions such as RAG remains underexplored, motivating our direct comparison.

3 Methods

3.1 Datasets

We evaluate on two medical QA benchmarks that represent different task types and hallucination detection modalities.

MedQA (USMLE) (Jin et al., 2021) consists of approximately 12,000 multiple-choice questions derived from United States Medical Licensing Examinations, covering a broad range of clinical topics. Gold-standard answer letters allow direct accuracy measurement. We evaluate on the held-out test split, sampling 450–500 items per experimental condition.

PubMedQA (Jin et al., 2019) consists of 450 expert-annotated yes/no/maybe questions derived from PubMed research abstracts. Each question is paired with the source abstract, making it ideal for evaluating context-grounded generation and retrieval augmentation. We use the full labeled training split (450 items) for all PubMedQA conditions.

3.2 Models

We compare three models spanning capability tiers:

Meta-Llama-3-8B-Instruct is a large general-purpose instruction-tuned model serving as our strong baseline. It was not fine-tuned on medical data.

BioMistral-7B (Labrak et al., 2024) is a medically-adapted model built on Mistral-7B and further pretrained on biomedical corpora. It allows us to test whether domain-specific pretraining reduces hallucination relative to a general model.

BioGPT (Luo et al., 2022) is a smaller (1.5B parameter) generative model pretrained on PubMed abstracts. It serves as both a small-model baseline and the target for our supervised fine-tuning experiments.

3.3 Experimental Conditions

We evaluate each model under the following conditions:

Zero-shot prompting provides the model with the question and answer choices (MedQA) or question alone (PubMedQA) with no additional context or reasoning instruction.

Chain-of-thought (CoT) prompting asks the model to reason step-by-step before providing a final answer, ending with `Answer: <letter>`.

Retrieval-augmented generation (RAG) uses BM25 retrieval over the PubMedQA abstract cor-

pus with top- $k=2$ abstracts prepended as context, using a fixed prompt template across all models.

Supervised fine-tuning (SFT) fine-tunes BioGPT on the full MedQA training split (10,178 samples) for 3 epochs using standard next-token prediction loss, learning rate 2×10^{-5} , effective batch size 16, and fp16 mixed precision on a single A100 GPU.

SFT + RAG applies RAG at inference time to the fine-tuned BioGPT model, testing whether the two interventions are complementary or redundant.

3.4 Evaluation Metrics

Accuracy is computed as the proportion of outputs whose extracted answer exactly matches the gold-standard answer. For MedQA, this means the extracted letter (A–D) matches the gold answer letter. For PubMedQA, this means the extracted decision (yes/no/maybe) matches the gold label.

Error rate is defined as $1 - \text{accuracy}$ over all outputs including parse failures. For MedQA conditions (zero-shot, CoT, SFT), we use error rate rather than “hallucination rate” since there is no retrieved context to assess faithfulness against — incorrect answers reflect factual errors rather than hallucination in the retrieval faithfulness sense.

Hallucination rate is reserved for RAG conditions on PubMedQA, where it captures outputs that are either incorrect or unfaithful to the retrieved context. Parse failures are included in the hallucination count since an unparseable output is clinically unusable.

Parse failure rate is reported separately as the proportion of outputs from which no structured answer could be extracted using regex pattern matching. For MedQA, the extractor looks for `Answer: [A–D]` or a leading letter. For PubMedQA, it looks for yes, no, or maybe at the start of or within the output. Parse failures are counted as errors in all conditions.

BERTScore F1 (Zhang et al., 2020) measures token-level semantic similarity between generated outputs and reference answers using contextual embeddings from RoBERTa-large. It serves as a secondary signal of output quality independent of exact-match accuracy.

4 Results

4.1 RQ1: Zero-Shot and Chain-of-Thought on MedQA

Table 1 summarises the performance of Meta-Llama-3-8B-Instruct on MedQA under zero-shot and CoT prompting.

Metric	Zero-Shot	CoT
Correct	302	282
Accuracy	60.4%	56.4%
Error Rate	39.6%	43.6%
Parse Failures	4	18
BERTScore F1	0.819	0.814

Table 1: MedQA results for Llama-3-8B-Instruct (500 samples).

Zero-shot prompting outperforms CoT on every metric. CoT produces longer reasoning traces that introduce more formatting artifacts, repetition, and parse failures, leading to a higher measured error rate. These results suggest that explicit reasoning prompts do not automatically improve performance on medical multiple-choice QA.

We observe a consistent pattern with Qwen2.5-3B-Instruct on MedQA (200 samples), as shown in Table 2.

Metric	Zero-Shot	CoT
Correct	86	47
Accuracy	43.0%	23.5%
Error Rate	57.0%	76.5%
Parse Failures	1	12

Table 2: MedQA results for Qwen2.5-3B-Instruct (200 samples).

Qwen2.5-3B-Instruct performs substantially worse than Llama-3-8B-Instruct under both conditions. The gap between zero-shot and CoT is even larger for Qwen2.5-3B, suggesting that smaller models are more susceptible to the noise introduced by chain-of-thought reasoning.

4.2 RQ2: Retrieval-Augmented Generation on PubMedQA

To directly evaluate whether RAG reduces hallucination, we ran both zero-shot and RAG conditions on the same 450 PubMedQA questions for all three models. Tables 3 and 4 report the results.

Model	Acc.	Halluc.	Parse Fail	BERT
Llama-3-8B	56.00%	44.00%	0.00%	0.830
BioMistral-7B	46.00%	54.00%	16.22%	0.838
BioGPT	22.89%	77.11%	57.56%	0.785

Table 3: PubMedQA zero-shot results (450 samples per model).

Model	Acc.	Halluc.	Parse Fail	BERT
Llama-3-8B	71.11%	28.89%	3.33%	0.853
BioMistral-7B	45.11%	54.89%	14.67%	0.820
BioGPT	10.89%	89.11%	79.78%	0.801

Table 4: PubMedQA RAG results with BM25 top-2 retrieval (450 samples per model).

The results reveal three distinct patterns across model tiers. For Llama-3-8B-Instruct, RAG substantially improves performance — accuracy increases from 56.00% to 71.11% (+15.11 points) and hallucination rate drops from 44.00% to 28.89%, with BERTScore F1 also improving from 0.830 to 0.853. For BioMistral-7B, RAG has essentially no effect — accuracy changes by only −0.89 points (46.00% → 45.11%), suggesting that domain-pretrained models already possess the parametric knowledge that retrieval provides and do not benefit from additional context. For BioGPT, RAG actively degrades performance — accuracy drops from 22.89% to 10.89% (−12 points) and parse failures increase from 57.56% to 79.78%, indicating that the retrieved context confuses the model rather than helping it generate structured responses.

4.3 RQ3: Effect of Fine-Tuning on Hallucination

We fine-tuned BioGPT on the full MedQA training split (10,178 samples) for 3 epochs. Training loss decreased steadily across epochs (2.05 → 1.73 → 1.63) with validation loss following the same trend (1.78 → 1.72 → 1.71), confirming convergence without overfitting. The fine-tuned model was evaluated on a matched subsample of 450 MedQA test items (seed=42) alongside the zero-shot baseline. Table 5 reports the results.

Condition	Accuracy	Error Rate	Parse Fail
BioGPT zero-shot	17.78%	82.22%	23.11%
BioGPT fine-tuned	24.44%	75.56%	0.00%

Table 5: RQ3: BioGPT on MedQA before and after fine-tuning (450 matched samples).

Fine-tuning improved accuracy by 6.66 percentage points and completely eliminated parse failures (23.11% → 0.00%), indicating the model learned to produce clean answer letters consistently. Despite this improvement, fine-tuned BioGPT (24.44%) remains substantially below zero-shot Llama-3-8B-Instruct (60.4%), suggesting that model scale and instruction-tuning capability are more important factors than domain-specific fine-tuning alone.

4.4 RQ4: RAG and Fine-Tuning Interaction

We applied RAG at inference time to the fine-tuned BioGPT model on PubMedQA to test whether the two interventions are complementary. Table 6 reports the results for three BioGPT conditions: RAG only (base model, PubMedQA), SFT only (fine-tuned, MedQA), and SFT + RAG (fine-tuned, PubMedQA with retrieval).

Condition	Acc.	Halluc.	Parse Fail
BioGPT RAG only	10.89%	89.11%	79.78%
BioGPT SFT only	24.44%	75.56%	0.0%
BioGPT SFT + RAG	3.33%	96.67%	93.33%

Table 6: RQ4: BioGPT under RAG, SFT, and combined conditions. RAG and SFT+RAG evaluated on PubMedQA (450 samples); SFT evaluated on MedQA (450 samples).

BioGPT struggles under all conditions on PubMedQA — zero-shot achieves only 22.89% (Table 3), RAG alone drops this further to 10.89%, and SFT + RAG collapses to 3.33% with a 93.33% parse failure rate. The consistent degradation across conditions suggests that BioGPT fundamentally lacks the capacity for structured yes/no/maybe output, independent of whether retrieval or fine-tuning is applied.

An important methodological caveat must be noted: the SFT model was fine-tuned on MedQA (multiple-choice, A/B/C/D format) and then evaluated on PubMedQA (yes/no/maybe format). These are fundamentally different output formats. The observed degradation in the SFT + RAG condition therefore reflects not only BioGPT’s limited capacity, but also a *cross-task format mismatch* — the fine-tuned model learned to produce answer letters and lost the output flexibility required for yes/no/maybe responses. This confounding effect limits the conclusions that can be drawn about the RAG + fine-tuning interaction specifically.

Additionally, PubMedQA contains only 450 labeled samples in total, all of which were used for

evaluation. This small dataset size limits statistical power and generalizability.

Despite these limitations, the result reveals a practically important finding: task-specific fine-tuning on a fixed output format introduces format rigidity that compounds existing model limitations, making the combined SFT + RAG approach worse than either intervention alone. A rigorous evaluation of the RAG + fine-tuning interaction would require either (1) fine-tuning on PubMedQA specifically and evaluating with and without RAG on a held-out split, or (2) using a larger dataset that supports both training and evaluation splits. We leave this for future work.

4.5 Prediction Distribution Analysis

Qwen2.5-3B-Instruct shows a strong bias toward hedging. On PubMedQA, 87.5% of its predictions were “maybe,” compared to a gold-label “maybe” rate of only 11.0%. On MedQA (zero-shot), the model exhibited a preference for option D (55.0% of predictions vs. 26.0% in the gold labels), while CoT prompting shifted the distribution heavily toward option A (60.5%). This sensitivity of the answer distribution to the prompting strategy suggests that the model’s predictions are driven more by surface-level prompt cues than by genuine medical reasoning.

5 Discussion

5.1 Effect of Model Scale

Across all conditions, model scale and instruction-tuning capability emerge as the dominant factors in hallucination reduction. Llama-3-8B-Instruct consistently outperforms both BioMistral-7B and BioGPT despite having no medical domain pre-training. This is consistent with recent findings that general large models with strong instruction tuning can match or exceed smaller specialized models on medical benchmarks (Singhal et al., 2023). Our results suggest that investing in larger, better-aligned general models may be more effective than domain-specific pretraining for reducing hallucination in medical QA.

5.2 Inference-Time vs. Training-Time Interventions

Our results reveal an important asymmetry between inference-time and training-time interventions. RAG substantially reduces hallucination for Llama-3-8B (+15.11 accuracy points, hallu-

cination drops from 44.00% to 28.89%) but has no measurable effect on BioMistral-7B (−0.89 points) and actively hurts BioGPT (−12 points). This suggests that RAG is only effective when the model has sufficient instruction-following capacity to leverage retrieved context. Fine-tuning improves BioGPT accuracy by 6.66 percentage points and completely eliminates parse failures (23.11% → 0.00%), demonstrating that task-specific adaptation is beneficial even for small models. However, the absolute accuracy of the fine-tuned model (24.44%) remains far below zero-shot Llama-3-8B (60.4%), suggesting that model scale and instruction-tuning capability are more critical factors than fine-tuning alone for hallucination reduction in medical QA.

5.3 Format Rigidity as a Hidden Risk of Fine-Tuning

A particularly important finding emerges from our RQ4 experiment. BioGPT struggles on PubMedQA under all conditions — zero-shot (22.89%), RAG (10.89%), and SFT + RAG (3.33%) — revealing that the model fundamentally lacks the capacity for structured yes/no/maybe output. Critically, applying SFT compounds this limitation: the model, having been trained exclusively on MedQA multiple-choice format, further loses output flexibility and achieves a 93.33% parse failure rate under SFT + RAG.

This finding has practical implications beyond our specific experimental setup: fine-tuning a model on a fixed output format introduces format rigidity that may prevent it from generalizing to other tasks at inference time, and can compound existing model weaknesses rather than mitigate them. Practitioners deploying fine-tuned models in multi-task clinical settings should carefully evaluate format compatibility before combining fine-tuning with retrieval augmentation.

We acknowledge an important caveat: our RQ4 setup involves a cross-task format mismatch, and PubMedQA’s 450 labeled samples are too few to support both a fine-tuning split and a reliable evaluation split. A fully controlled interaction test — fine-tuning and RAG evaluation within the same format and dataset — remains for future work. Nevertheless, the finding that SFT compounds rather than alleviates BioGPT’s structured output failures stands independently of this limitation.

6 Conclusion

We presented a controlled empirical study of hallucination behavior in medical LLMs across four intervention conditions. Our key findings are: (1) larger instruction-tuned models hallucinate less under RAG, with Llama-3-8B achieving 71.11% accuracy on PubMedQA; (2) CoT prompting consistently degrades performance relative to zero-shot due to output noise and parse failures; (3) fine-tuning BioGPT improves accuracy by 6.66 percentage points and completely eliminates parse failures, though absolute accuracy remains bounded by model scale; and (4) combining fine-tuning with RAG reveals a format rigidity problem — task-specific fine-tuning can severely degrade output format flexibility, preventing generalization to new task formats even with retrieved context. These findings highlight that model scale and instruction-tuning capability are more important than domain-specific adaptation for hallucination reduction, and that combined interventions must be carefully evaluated for format compatibility rather than assumed to be additive.

7 Limitations

Several limitations of this work should be noted. First, our RQ4 evaluation involves a cross-task format mismatch: BioGPT was fine-tuned on MedQA (multiple-choice, A/B/C/D) but evaluated with RAG on PubMedQA (yes/no/maybe). A rigorous interaction test would require fine-tuning and evaluation within the same format and dataset. PubMedQA’s 450 labeled samples are too few to support both a fine-tuning split and a reliable evaluation split simultaneously, which is why we could not conduct this experiment within the current study.

Second, our fine-tuning used only 3 epochs with standard SFT; longer training or alternative objectives such as RLHF or DPO may yield greater hallucination reductions. Finally, we use BM25 retrieval with top-2 abstracts; denser retrieval methods such as MedCPT embeddings may improve RAG quality and faithfulness.

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